
Strings that tie: Network modeling and interactive visualization of musical instrument makers in Valdrighi's *Nomocheliurgografia*

Federico Filippi Prévost De Bord*¹

¹University of Genoa (UniGe) – Italy

Abstract

The increasing availability of digitized historical sources has gradually shifted attention from simple preservation towards the creation of exploratory research environments. In this context, network analysis approaches encourage the rethinking of large corpora not only as collections of information, but also as systems of relationships that support alternative forms of interpretation. This study presents the workflow adopted to transform Luigi Francesco Valdrighi's three-volume expanded edition of *Nomocheliurgografia* (1884, 1888, 1894), an extensive biographical dictionary of European musical instrument makers, into an interactive network environment. While Valdrighi's encyclopaedic compilation is still one of the most important sources for the history of craftsmanship from the 19th-century, its linear structure makes it difficult to perceive broader relational patterns among the entries (Tamper et al. 2023). The idea behind the process is not limited to the creation of visual summaries, but also serves as a heuristic device for research (Ahnert et al. 2021, 92).

The development of a custom routine has facilitated the conversion of the book's pages into structured, analyzable datasets. Starting from digitized page images, the workflow uses a multimodal vision-language model to recognize the tabular structure and convert it into CSV records. The output is then cleaned and normalized in Python in order to construct a bipartite graph in NetworkX (Ladd et al. 2017). Particular attention has been devoted to the stabilization of historically inconsistent data, including spelling variants, OCR noise, and geographically fragmented references, highlighting the interpretative dimension involved in the transformation of historical sources into a structured dataset. Individuals are then connected with their associated attributes (nationality, city, musical instrument, school or style) (Figure 1), while preserving the possibility of consulting single biographies through direct name searches, similarly to a traditional dictionary (Figure 2). In order to provide an interactive web interface, the graph is then rendered through Dash-Cytoscape. Combining a dropdown menu, year-range slider and a text-based filter, the user triggers a callback function which rebuilds and updates the visualization in real time.

Preliminary observations on the letters A to C draw attention to strong internal cohesion regarding certain instrument-making traditions, while others form clusters that connect distant geographical areas, providing an alternative perspective to traditional taxonomies. Moreover, temporal filtering shows shifts in centres of craftsmanship activity over the centuries, emphasizing mobility and the transmission of workshop practices. Current work focuses on experimenting and refining the workflow to re-read other historical sources through dynamic relational lenses, creating custom heuristic tools for musicology.

*Speaker

Bibliography

Valdrighi, Luigi Francesco. 1884, 1888, 1894. *Nomocheliurgografia antica e moderna ossia elenco di fabbricatori di strumenti armonici con note esplicative e documenti estratti dall'Archivio di Stato di Modena*. Antica Tipografia Soliani.

Ahnert, Ruth, Sebastian Ahnert, Catherine Coleman, and Scott Weingart. 2021. *The Network Turn: Changing Perspectives in the Humanities*. Cambridge University Press.

Ladd, John, Jessica Otis, Christopher Warren, and Scott Weingart. 2017. "Exploring and Analyzing Network Data with Python." *Programming Historian* 6. <https://doi.org/10.46430/phen0064>.

Tamper, Minna, Petri Leskinen, and Eero Hyvönen. 2023. "Visualizing and Analyzing Networks of Named Entities in Biographical Dictionaries for Digital Humanities Research." *Computational Linguistics and Intelligent Text Processing*, edited by Alexander Gebukh, 199-214. Lecture Notes in Computer Science. Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-24337-0_15.

Keywords: Digital Musicology, Biographical dictionaries, Multimodal data extraction, Network visualization, Organology

Strings that Tie: Network Modeling and Interactive Visualization of Musical Instrument Makers in Valdrighi's *Nomocheliurgografia*

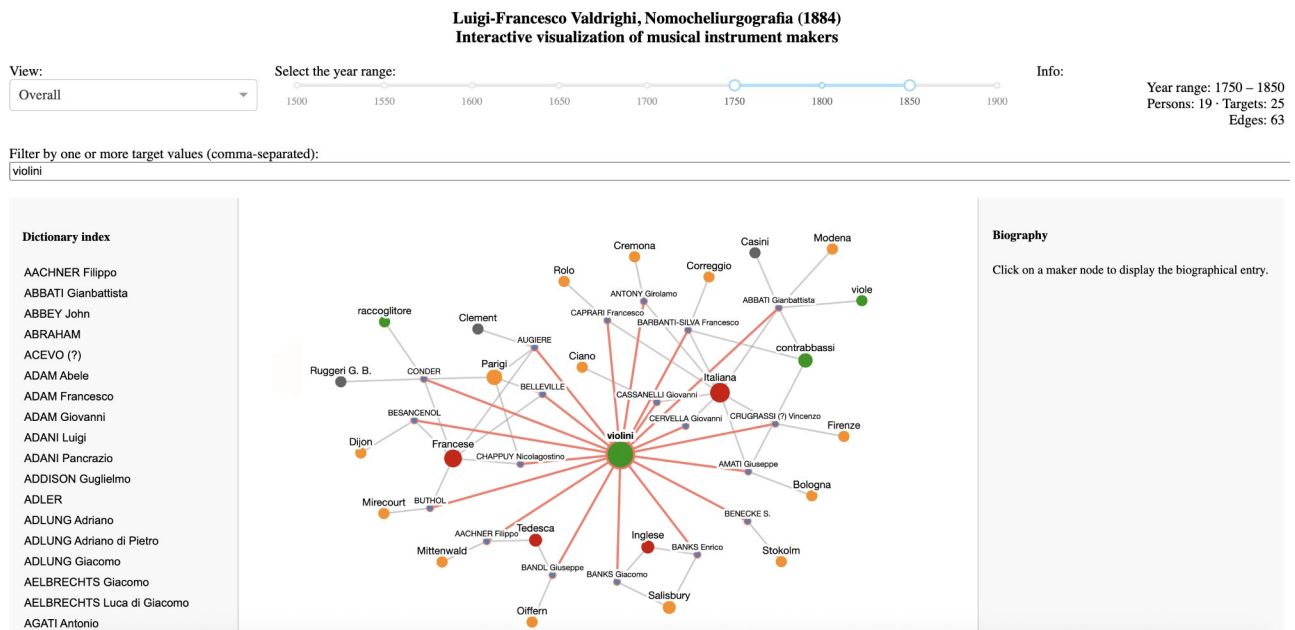


Figure 1. Overall view example showing makers (blue), nationalities (red), cities (yellow), musical instruments (green), and schools (grey) for the letters A-C, filtered by the search term “violini” and the year range 1750-1850.

Strings that Tie: Network Modeling and Interactive Visualization of Musical Instrument Makers in Valdrighi's *Nomocheliurgografia*

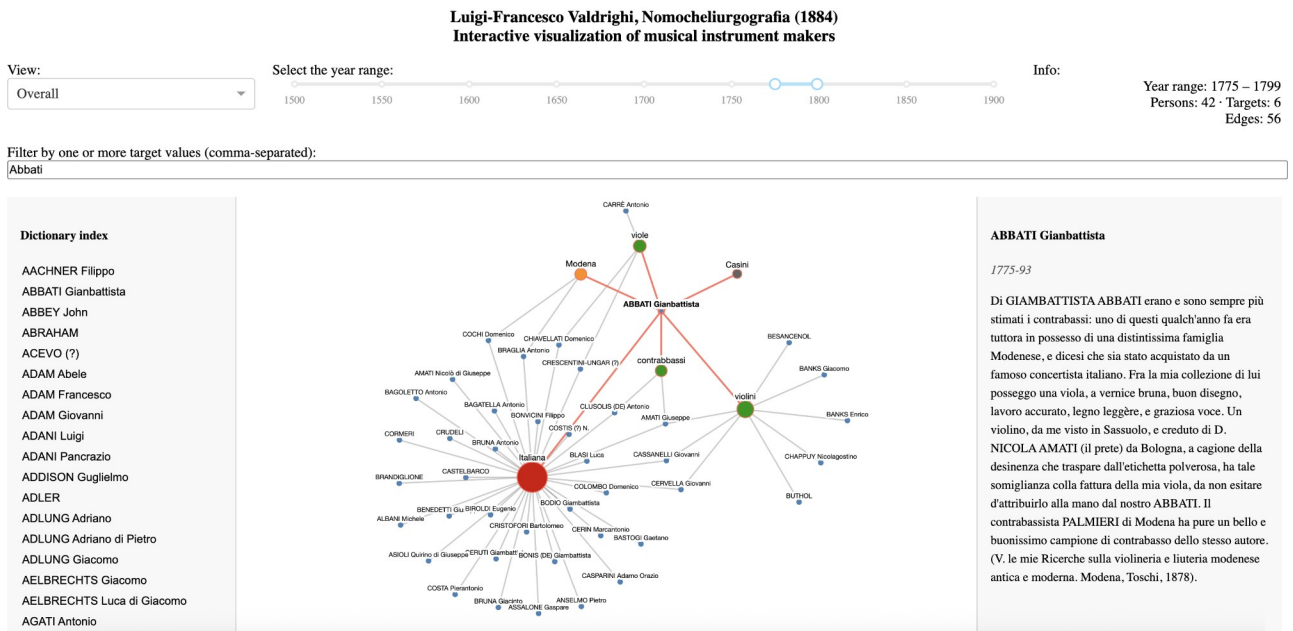


Figure 2. Focus on an individual maker, showing the highlighted node, associated attributes, and biographical panel.